

The Multiplier Group of a Singer Difference Set is not the Norm Subgroup:

Disproof of Conjecture 00000001031

Conjecture 00000001031. *Let $D \subset \mathbb{F}_{q^3}$ be a Singer difference set, and let its multiplier group be the set of multiplicative translations τ with $\tau D = D$. Then the multiplier group is exactly the norm subgroup $N(\mathbb{F}_{q^3}/\mathbb{F}_q)$ of $\mathbb{F}_{q^3}^*$, of index $q^2 + q + 1$; this structure completely characterizes the automorphisms of Singer difference sets, with no other multipliers.*

Theorem. *The conjecture is **false**. It already fails for $q = 2$.*

1 The counterexample

Work in $\mathbb{F}_8 = \mathbb{F}_2[\alpha]$ with $\alpha^3 = \alpha + 1$, so that $\mathbb{F}_8^* = \langle \alpha \rangle$ is cyclic of order 7; write $\alpha^k \leftrightarrow k \in \mathbb{Z}_7$. The classical Singer difference set is the set of nonzero trace-zero elements,

$$D = \{x \neq 0 : \text{Tr}_{8/2}(x) = x + x^2 + x^4 = 0\} = \{1, 2, 4\} \subset \mathbb{Z}_7.$$

Indeed the six ordered differences of distinct elements of D are 6, 4, 1, 5, 3, 2, i.e. every nonzero residue of $\mathbb{Z}_7$ occurs exactly once, so D is a $(7, 3, 1)$ difference set with $\lambda = q - 1$.

Proposition (Multiplier group). *The multipliers of D in $\mathbb{Z}_7$ are exactly $\{1, 2, 4\}$, a group of order 3.*

Proof. Direct check for all units $\tau \in \mathbb{Z}_7^* = \{1, \dots, 6\}$:

$$\begin{aligned} 1 \cdot D &= \{1, 2, 4\} = D & 2 \cdot D &= \{2, 4, 1\} = D & 3 \cdot D &= \{3, 6, 5\} \neq D, \\ 4 \cdot D &= \{4, 1, 2\} = D & 5 \cdot D &= \{5, 3, 6\} \neq D & 6 \cdot D &= \{6, 5, 3\} \neq D. \end{aligned}$$

Hence the multiplier group is $\{1, 2, 4\} = \langle 2 \bmod 7 \rangle$, the subgroup generated by the Frobenius $x \mapsto x^2$. □

Proposition (Norm subgroup). *The norm map $N : \mathbb{F}_8^* \rightarrow \mathbb{F}_2^*$, $N(x) = x^{1+2+4} = x^7$, has image $N(\mathbb{F}_8/\mathbb{F}_2) = \mathbb{F}_2^* = \{1\}$: a subgroup of order $q - 1 = 1$ and index $7 = q^2 + q + 1$ in $\mathbb{F}_8^*$.*

Theorem. *The multiplier group $\{1, 2, 4\}$ of D has order 3, while the conjectured norm subgroup $\{1\}$ has order $q - 1 = 1$. Since $\{1, 2, 4\} \neq \{1\}$, the conjectured equality fails for $q = 2$. Moreover the clause “no other multipliers” fails too: the multipliers 2 and 4 do not lie in the norm subgroup.*

2 Boundary and remarks

Only $q = 2$ is computed; this suffices, as the conjecture is a universal “exactly” statement. Nothing is claimed for general q . The computation is reproduced by the standalone script `reproduce.py` (polynomial arithmetic in $\mathbb{F}_8$ from scratch) and machine-checked in core Lean 4 in the directory `lean4/` (zero axioms, zero `sorry`; run `lake build && lake env lean Check.lean`).